
Specialist Physical AI Need Not Be Opaque A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

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Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ -VLA, a rational vision-language-action model family with a 19.2M convolutional policy and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned operations. Their weights can therefore be reorganized into explicit interaction tensors rather than treated only as an opaque checkpoint. A mechanical audit and numerical reconstructions verify that the deployed policy preserves this form. Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches $93.7\% \pm 0.8\%$ over three seeds under the published protocol. The three released convolutional checkpoint files associated with those historical aggregates also pass an independent 4,608-case joint-attention certificate with worst relative error 6.36×10^{-16} . In a separate ViT instantiation, we reconstruct every attention module in three checkpoints with worst relative error 2.56×10^{-7} , and exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. The result is a capable and structurally inspectable specialist family with exact access to trained attention coefficients. It does not yet provide an exact end-to-end decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

34 We study the narrower but testable question:

35 *Can a specialist VLA retain strong closed-loop capability when every deployed*
 36 *learned operation is polynomial or rational, while supporting exact weight analy-*
 37 *sis?*

38 We test capability and structural analysis in two fully rational encoder instantiations. The convo-
 39 lutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact
 40 attention audit.

41 Our contributions are:

- 42 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
 43 supplies input-specific scaling without leaving the rational function class.
- 44 2. We evaluate fully rational 19.2M convolutional and 20.1M ViT policies under a protocol
 45 aligned with published LIBERO-Object results. The strongest single architecture reaches
 46 $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
- 47 3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational
 48 and bilinear branches from their weights.
- 49 4. We extend exact weight-derived decomposition through individual attention layers with a
 50 reduced-Gram calculation that avoids materializing the full high-order tensor.

51 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 52 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 53 policy makes exact weight structure directly accessible.

54 2 A rational tensor robot policy

55 2.1 Tensor-compatible primitives

56 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 57 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 58 interactions that remain determined by the weights.

59 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 60 layer to represent bias, linear, and quadratic terms in one contraction.

61 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

62 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 63 CP factorization, the tensor analogue of a low-rank matrix factorization.

64 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^T) \odot (Q_2 K_2^T) / s] V. \quad (2)$$

65 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
 66 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
 67 interaction coefficients determined by the weights.

68 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 69 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
 70 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 4. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

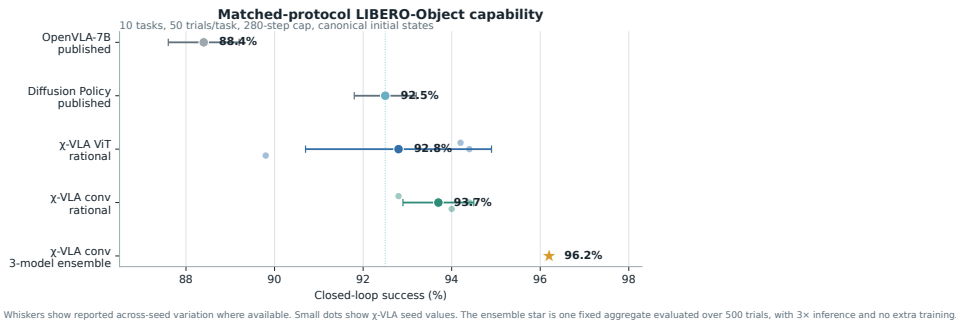


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

101 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 19.2\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

102 The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow
103 spread is descriptive bounded evidence across independently trained checkpoints.

104 Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and
105 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points
106 above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato
107 sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one
108 19.2M-parameter policy.

109 3.3 What this result does and does not establish

110 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
111 existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization
112 or state of the art.

113 4 Exact weight-derived structure through attention

114 4.1 Exact layerwise construction

115 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
116 a coefficient table whose entries measure how combinations of query, key, and value coordinates
117 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
118 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
119 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

120 The deployed attention also normalizes each query and key with Equation 3. Each normalization
121 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
122 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
123 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
124 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
125 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
126 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
127 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

128 We separately certify the three released convolutional checkpoint files associated with the historical
129 capability aggregates. Their vision stacks contain no attention, so we independently rebuild all eight
130 joint modules and all twelve heads per module from the saved weights. Sixteen deterministically
131 selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every
132 output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This
133 same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does
134 not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or
135 action head as one end-to-end contraction.

136 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
137 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
138 rank important directions without constructing the full table. It matches brute-force materialization to
139 2.13×10^{-14} at toy scale.

140 4.2 Trained policy result

141 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
142 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
143 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
144 activations are used only afterward to score how well it preserves the layer’s computation.

145 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
146 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
147 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
148 0, 6, and 7.

149 This extends exact weight-only construction through individual attention layers. It is not an exact
150 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
151 rapid degree and representation growth.

152 5 Related work and limitations

153 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
154 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
155 weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also
156 steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived
157 construction in a policy designed as a rational tensor program. Tensor and polynomial networks
158 provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard
159 LIBERO success can coexist with weak language use [9–11].

160 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
161 robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce
162 one compact symbolic contraction of the full policy. The released convolutional files support the
163 joint-attention equation certificate and are associated with the historical capability aggregates. The
164 reduced-Gram subspace screen uses separate ViT checkpoints. Evaluation uses fixed task instructions
165 and does not establish robust instruction grounding. Within that scope, rational restrictions support
166 strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct
167 coefficient-edit interface.

168 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
169 certificates, and per-episode logs. Historical Modal cache-backed analyses read dataset-native task
170 metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official
171 identifiers before defining task splits. A source-level audit compared all 271,996 cached frames
172 across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers,
173 language joins, states, actions, and resized images matched exactly for every frame. Canonical-state
174 checks fail loudly when requested states are unavailable.

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200 A Supporting results and decisions

201 A.1 Per-task ensemble behavior

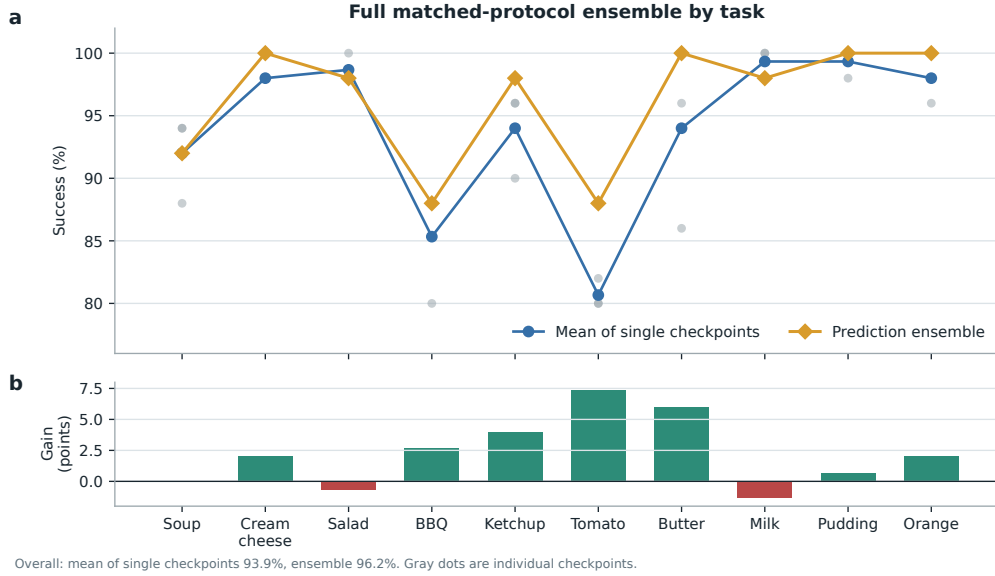


Figure 2: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

202 A.2 Expanded attention screen

203 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 204 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 205 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 206 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 207 tested rank.

208 A.3 Numerical scope of rational conversion

209 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 210 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 211 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 212 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 213 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 214 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 215 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 216 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 217 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 218 gate.

219 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 220 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 221 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 222 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 223 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 224 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 225 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

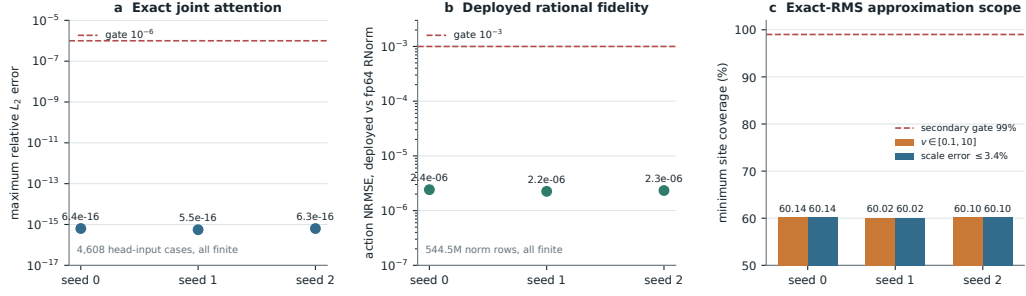


Figure 3: **Frozen structural certificates on the released convolutional files associated with the historical capability aggregates.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.

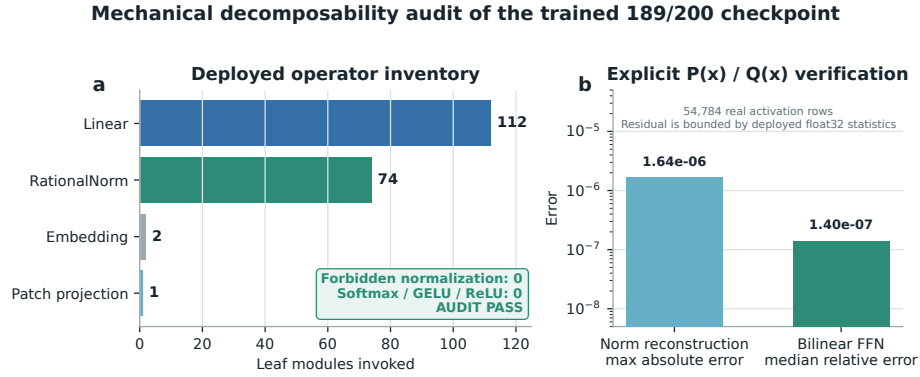


Figure 4: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.